

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`

D. `<vid src="video.mp4"></vid>`

Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

Snippet 3: Navigation Elements

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

Snippet 4: XML Syntax

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`
- D. `<note to="User"></note>`

Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I B.Maheshwar (192411195) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Maheshwar

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

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Assessment tool - 3

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: Option B: `<video src="video.mp4" controls></video>`

Explanation:

- The `<video>` tag is the standard HTML5 element used to embed video files directly into a web page.
- The 'controls' attribute automatically provides Play, Pause, Volume, Progress Bar, and Full Screen buttons, making the video easy for users to control.
- It supports multiple video formats such as MP4, WebM, and OGG when used with the `<source>` element.
- The tag is supported by all modern web browsers and does not require third-party plugins like Adobe Flash.
- The other options (`<media>`, `<movie>`, and `<vid>`) are not valid HTML5 elements and therefore will not work correctly in browsers.

2. Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: Option B: POST

Explanation:

- POST is the recommended HTTP method for registration and login forms because it sends the data in the request body instead of the URL.
- This makes sensitive information such as passwords, phone numbers, and email addresses more secure than GET requests.
- POST allows large amounts of data to be transmitted to the server without URL length limitations.
- It is commonly used for creating new records in databases, such as user registration.
- GET is mainly used to retrieve information, while PUT updates existing data and DELETE removes data from the server.

3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: Option B: `<nav>`

Explanation:

- The `<nav>` element is a semantic HTML5 tag specifically designed for navigation menus.

- It clearly identifies the section containing links to important pages such as Home, About, Services, and Contact.
- Semantic elements improve the readability of HTML code and make websites easier to maintain.
- They also improve accessibility because screen readers can easily identify navigation sections for visually impaired users.
- Search engines use semantic tags to better understand the structure of a webpage, improving SEO.

4. Identify the correct XML syntax and explain the error in other options.

Correct Answer: Option B: `<note><to>User</to><from>Admin</from></note>`

Explanation:

- XML requires every opening tag to have a matching closing tag and all elements must be properly nested.
- Option B follows all XML rules because every element is correctly opened and closed.
- Option A is incorrect because the closing `</note>` tag is missing.
- Option C is incorrect because the opening `<to>` tag is closed using `</from>`, which causes mismatched tags.
- Option D is incorrect because the final angle bracket (`>`) is missing, making the XML document invalid.

5. Choose the correct protocol for web communication and justify your choice.

Correct Answer: Option B: HTTP (HyperText Transfer Protocol)

Explanation:

- HTTP is the standard communication protocol used between web browsers (clients) and web servers.
- It enables users to request web pages, images, videos, and other online resources from a server.
- When a user enters a website address, the browser sends an HTTP request and the server returns an HTTP response.
- Most modern websites use HTTPS, which is the secure version of HTTP that encrypts transmitted data using SSL/TLS.
- The other protocols have different purposes: FTP transfers files, SMTP sends emails, and SNMP manages network devices.

Conclusion

The startup should use the HTML5 `<video>` tag for multimedia, POST for secure form submission, the

<nav> semantic element for navigation, valid XML syntax with proper opening and closing tags, and HTTP/HTTPS for communication between the client and the web server. Following these web standards ensures security, compatibility, accessibility, and reliable web application performance.